



RPA Device Pools & Administration

PPT PRESENTED BY

NAME : Srinivas Rao Tammireddy

ROLL NO : 257Y1D5805

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Device Pools & Runtime Clients

Device Pools group multiple runtime client machines (bot runners) into clusters. They allow the Web Control Room to allocate tasks dynamically, providing scalability and high availability.

Managing Device Pools

- **Runtime Clients:** Host machines that run compiled bots. Classified into unattended (automated) and attended (user-triggered).
- **Clustering:** Grouping multiple runners into a device pool. If runner A is busy, Web CR routes tasks to runner B.
- **High Availability:** If a runner machine crashes, the pool automatically redirects the queue to active runner nodes.
- **Dynamic Scaling:** Admins can add or remove runner nodes to the pool to scale capacity.

Workloads & Queue Management

- **Transaction Queues:** Lists of data records (e.g. 5,000 invoices) to be processed by a bot.
- **Workload Distributor:** Web CR reads items from the queue and distributes them across device pool runners concurrently.
- **SLA Calculator:** Calculates the estimated time to complete a queue based on processing speeds and node counts. Alerts admins if SLAs are at risk.
- **Queue Security:** Encrypts data elements inside database queues.



RPA Administration & Roles

Administration tools secure, audit, and configure the enterprise RPA platform:

User & Role Management

- **Role-Based Access (RBAC):** Admins configure roles with specific permissions, enforcing security rules:
 - **Bot Creator:** Permissions to write scripts and edit local code. Cannot deploy to production.
 - **Bot Runner:** Permissions to run scripts. Cannot edit code.
 - **Control Room Admin:** Manages users, devices, licenses, and system configurations.
- **License Allocation:** Allocates developer and runtime runner licenses.

Audit Log Compliance

- **Security Compliance Audits:** Logs all activities in the Web CR. Cannot be edited by users or admins.
- **Actions Tracked:**
 - User modifications (role updates, password changes).
 - Bot modifications (uploads, deletions, runs).
 - Device changes (pool creation, connection edits).
- **Compliance value:** Essential for SOX, HIPAA, and GDPR database audits.

Applications of RPA Administration

How organizations deploy device pooling and role segregation in production:



Scaling for Seasonal Invoices

During holiday sales, invoice volumes multiply. Admins add runtime servers to the billing device pool, allowing parallel processing and meeting SLAs.



Compliance Role Segregation

Enforces security rules. Developers create bots but have no runner permissions in production. Operations staff run bots but cannot edit scripts.



Auditing Financial Bot Runs

Auditors review Web CR audit logs to confirm only authorized bots executed bank transfers, matching transaction timestamps.



System Migration & Updates

RPA administrators manage migrations (e.g. importing bot packages from dev to staging schemas) and verify licensing compliance.



THANK YOU

Department of Computer Science and Engineering
Marri Laxman Reddy Institute of Technology and Management